

Hardy–Weinberg Equilibrium

Subject & topics: Biology | Evolution | Theory & Evidence Stage & difficulty: A Level P3

The Hardy–Weinberg principle states that, given certain conditions, genotype frequencies will remain constant at

$$p^2 + 2pq + q^2 = 1$$

where

- p^2 is the proportion of homozygotes for the allele for which the proportion is given by p
- $2pq$ is the proportion of heterozygotes in the population
- q^2 is the proportion of homozygotes for the allele for which the proportion is given by q

Note that $p + q = 1$.

Part A Equal proportions

A gene has two alleles (**A**, **a**) in a population. The proportion of allele **A** in the population (given by p) is 0.5.

Find q , the proportion of allele **a** in this population.

Assuming the population is in Hardy–Weinberg equilibrium, what percentage of individuals will be homozygous for allele **A**?

Part B

Unequal proportions

A gene has two alleles (**A**, **a**) in a population. The proportion of allele **A** in the population (given by p) is 0.8.

Find q , the proportion of allele **a** in this population.

Assuming the population is in Hardy–Weinberg equilibrium, what percentage of individuals will be homozygous for allele **a**?

Part C

How many heterozygotes?

A gene has two alleles (**A**, **a**) in a population of 2 000 individuals.

65% of the copies of this gene in this population are allele **A**.

How many individuals would you expect to be heterozygous in this population?

Part D

Disequilibrium

A gene has two alleles (**A**, **a**) in a population. The proportion of allele **A** in the population (given by p) is 0.5.

The genotype frequencies are as follows:

- **AA**: 45% of individuals
- **Aa**: 10% of individuals
- **aa**: 45% of individuals

Which of the following reasons may explain why this population is not in Hardy-Weinberg equilibrium for this gene?
Select all that apply.

- ☐ The population is very large and so genotype frequencies are highly affected by genetic drift.
- ☐ Recent immigration into the population has occurred.
- ☐ The population is very small and so genotype frequencies are highly affected by genetic drift.
- ☐ Homozygotes prefer to mate with individuals of the opposite homozygote genotype.
- ☐ Heterozygotes have a higher fitness than homozygotes.
- ☐ Recent emigration out of the population has occurred.
- ☐ Heterozygotes have a lower fitness than homozygotes.
- ☐ Homozygotes prefer to mate with individuals of the same genotype.

Created for isaacphysics.org by Lewis Thomson

Question deck:

STEM SMART Biology Week 29 - Hardy-Weinberg Equilibrium

Albino Rabbits

Subject & topics: Biology | Evolution | Theory & Evidence **Stage & difficulty:** A Level P3

The Hardy–Weinberg principle, represented by the equations below, can be used to estimate the frequency of alleles in a population.

$$p^2 + 2pq + q^2 = 1$$

$$p + q = 1$$

Albino rabbits have white fur as these individuals are unable to produce the pigment melanin. The ability to produce melanin is controlled by a gene with a dominant allele (**B**), resulting in brown fur, and a recessive allele (**b**), resulting in an albino.

Of the 60 rabbits in a pet shop, 45 are brown.

A student decided to use the Hardy–Weinberg principle to estimate the frequencies of the alleles in this group of rabbits.

Part A

Hardy–Weinberg calculation

Using the Hardy–Weinberg equations, estimate the frequency of the **dominant** allele in this group.

Give your answer to 2 decimal places.

Part B

Hardy–Weinberg conditions

Why was it not appropriate to use the Hardy–Weinberg principle to estimate the frequencies of alleles in this group of rabbits in the pet shop? Select all that apply.

- ☐ natural selection is likely to be occurring
- ☐ rabbits will regularly be leaving the pet shop (and new rabbits will be brought in)
- ☐ mating is likely to be random
- ☐ the number of rabbits is very small
- ☐ albinism is a mutation
- ☐ mating is **not** likely to be random

Adapted with permission from OCR A Level Biology A, June 2014, Control, Genomes and Environment, Question 6b

Question deck:

STEM SMART Biology Week 29 - Hardy-Weinberg Equilibrium

Budgie Genotypes

Subject & topics: Biology | Evolution | Theory & Evidence Stage & difficulty: A Level P3

The Hardy-Weinberg principle, represented by the equations below, can be used to estimate the frequency of alleles and genotypes in a population.

$$p^2 + 2pq + q^2 = 1$$

$$p + q = 1$$

A breeder of birds keeps a population of 86 budgerigars in one enclosed area. Two distinct phenotypes are present, blue feathers and green feathers. Feather colour is controlled by one gene:

- **G** is the allele for green feathers
- **g** is the allele for blue feathers

Only 17 of the budgerigars have blue feathers.

Part A

Heterozygous individuals

Estimate the number of **heterozygous** individuals in the population.

Part B

Homozygous dominant individuals

Estimate the number of **homozygous dominant** individuals in the population.

Hardy–Weinberg conditions

The Hardy–Weinberg principle does not apply to all populations.

Which of the following are conditions in which the Hardy–Weinberg principle does **not** apply?

- ☐ the population size is extremely small
- ☐ the population size is extremely large
- ☐ mating is random
- ☐ mating is non-random
- ☐ one allele is more common than the other
- ☐ one allele has a selective advantage over the other
- ☐ individuals are migrating into the population
- ☐ individuals are migrating out of the population

Adapted with permission from CIE A Level Biology, June 2018, Paper 4, Question 2c

Question deck:

STEM SMART Biology Week 29 - Hardy-Weinberg Equilibrium

Flour Beetle Eye Colour

Subject & topics: Biology | Evolution | Theory & Evidence **Stage & difficulty:** A Level P3

In flour beetles one gene controlling eye colour is located on chromosome 5. Flour beetles have two copies of chromosome 5 in each cell. One allele causes black eyes and a second allele causes red eyes.

The allele for black eye (B) is dominant over the allele for red eye (b).

$\frac{3}{4}$ of the alleles present in a population of 1600 flour beetles were the dominant B allele.

Part A

Black eyes

What is the expected number of flour beetles with black eyes?

Part B

Expected ratio of genotypes

What is the expected ratio of homozygous black eye beetles to heterozygous black eye beetles to red eye beetles?
Express your answer as a ratio in its simplest form (e.g. 1 : 2 : 3)

Adapted with permission from NSAA 2022 Specimen Paper Section 2 Q30

Plant Heights & Hardy–Weinberg Equilibrium

Subject & topics: Biology | Evolution | Theory & Evidence **Stage & difficulty:** A Level P3

In a particular species of plant, a gene that affects plant height has two codominant alleles (**T** and **t**).

- **TT** plants are tall.
- **Tt** plants are medium-height.
- **tt** plants are short.

A researcher counts the number of plants of each type in a population of 200 diploid plants. The results are shown in the table below.

Plant phenotype	Observed frequency
Tall	12
Medium-height	136
Short	52

The researcher wants to investigate whether the population is in Hardy–Weinberg equilibrium for this gene.

Part A

Allele proportions

State the **allele** proportions below, where

- p is the proportion of the **T** (tall) allele
- q is the proportion of the **t** (short) allele

Give your answers as exact decimals.

$p =$

$q =$

Part B

Expected frequencies

Using your answers to the previous section, fill in the expected frequencies (assuming Hardy-Weinberg equilibrium) in the table below.

Plant phenotype	Observed frequency	Expected frequency
Tall	12	
Medium-height	136	
Short	52	

Part C

Conclusion

We can see that the observed frequencies differ from the expected frequencies by a large amount. This suggests that this plant population is **not** in Hardy-Weinberg equilibrium for this gene.

Which of the following may explain why this population is not in Hardy-Weinberg equilibrium for this gene? Select all that apply.

- ☐ Heterozygotes have a **higher** fitness than homozygotes.
- ☐ Heterozygotes have a **lower** fitness than homozygotes.
- ☐ The population is small and therefore allele/genotype frequencies are highly affected by genetic drift.
- ☐ Tall plants cannot successfully reproduce with short plants.

ABO Blood Types & Hardy–Weinberg Equilibrium

Subject & topics: Biology | Evolution | Theory & Evidence **Stage & difficulty:** A Level C3

An individual's blood type is determined by two genes: the ABO gene and the RhD gene.

The ABO gene has three main alleles: I^A , I^B , and i (alternatively named I^O). I^A and I^B are codominant, and both are dominant to i .

A genomics study was carried out to measure the relative frequencies of different ABO genotypes in the UK. The genomes of 487 236 individuals were sequenced. The results are shown in the table below.

Blood type	Genotype	Frequency
A	$I^A I^A$	36 326
	$I^A i$	175 134
B	$I^B I^B$	2 787
	$I^B i$	44 026
AB	$I^A I^B$	17 606
O	ii	211 357

Data from <https://biobank.ctsu.ox.ac.uk/ukb/field.cgi?id=23165>

Answer the questions below to determine whether, based on this sample, the UK population is in Hardy–Weinberg equilibrium for the ABO gene.

Part A

Three alleles

For a gene with two alleles, where the proportion of one allele is given by p and the proportion of the other allele is given by q , the expected genotype proportions in Hardy–Weinberg equilibrium are:

$$p^2 + 2pq + q^2$$

Fill in the expected genotype proportions in Hardy–Weinberg equilibrium for the ABO gene, where

- p is the proportion of the I^A allele
- q is the proportion of the I^B allele
- r is the proportion of the i allele

Blood type	Genotype	Expected genotype proportion
A	$I^A I^A$	
	$I^A i$	
B	$I^B I^B$	
	$I^B i$	
AB	$I^A I^B$	
O	ii	

Items:

p^2

p^3

q^2

q^3

r^2

r^3

$2pq$

$3pq$

$2pr$

$3pr$

$2qr$

$3qr$

$2pqr$

$3pqr$

Part B

Allele proportions

State the allele proportions below, where

- p is the proportion of the I^A allele
- q is the proportion of the I^B allele
- r is the proportion of the i allele

Give your answers to 4 sf.

$p =$

$q =$

$r =$

Part C

Expected frequencies

Using your answers in the previous parts, calculated the expected frequency of each genotype in this dataset. You may use your rounded answers from part B. Give your answers to 3 sf.

Blood type	Genotype	Observed frequency	Expected frequency
A	$I^A I^A$	36 326	
	$I^A i$	175 134	
B	$I^B I^B$	2 787	
	$I^B i$	44 026	
AB	$I^A I^B$	17 606	
O	ii	211 357	

Part D

Conclusion

We can see that the observed frequencies only differ from the expected frequencies by a few hundred individuals. Given the large size of our sample (487 236), this is quite a small difference. This suggests that the UK population **is** in Hardy–Weinberg equilibrium for the ABO gene.

Based on this, which of the following conclusions are supported? Select all that apply.

- ☐ Individuals with the **ii** genotype have a higher fitness than individuals with any other genotype.
- ☐ There is **strong** selection acting on these alleles/genotypes.
- ☐ There is **no** selection acting on these alleles/genotypes.
- ☐ Individuals prefer to have children with individuals of a different blood type to themselves.
- ☐ Individuals do **not** show mating preferences based on blood type.

Created for isaacscience.org by Lewis Thomson. This question has been established using the UK Biobank Data Field 23165.